3D Design Studio

A project guide: what it is, how to run it, and how to publish it on GitHub

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Includes: web edition (three.js), Python edition (matplotlib), local setup, and a step-by-step GitHub publishing workflow.

Overview

This project contains two implementations of the same idea: a small 3D scene builder where you place primitive shapes, move and recolor them, and view the result from any angle.

Web edition	**Python edition**
Single HTML file built with three.js. Opens directly in any browser — no install, no server. Includes a drag gizmo for move/rotate/scale, a color palette, and a numeric position panel.	A plain Python script using matplotlib. Define a scene as a list of shapes and positions in code, then render an interactive, rotatable 3D view.

Features

- Primitive shapes — boxes, spheres, cylinders, cones, torus, capsules
- Transform tools — move, rotate, and scale (web edition gizmo)
- Live recoloring from a curated palette
- Numeric position control via an inspector panel (web edition)
- Free orbit camera — rotate, pan, and zoom
- Zero setup for the web edition — open the HTML file directly

Tech stack

Web edition	**Python edition**
three.js (r128) OrbitControls + TransformControls Plain HTML / CSS / JavaScript	matplotlib (mplot3d) NumPy No GUI framework required

Running it locally

Web edition

No install required. Double-click 3d-design-studio.html, or drag it into any modern browser window.

```
open 3d-design-studio.html
```

Python edition

Install the two dependencies, then run the script:

```
pip install matplotlib numpy
python 3d_design_app.py
```

Edit the demo_scene() function in the script to add your own shapes and positions.

Publishing to GitHub

These steps assume git is already installed and you have a free GitHub account.

1. Create a repository

Go to github.com, click the + icon top right, then New repository. Give it a name (e.g. 3d-design-studio), choose Public or Private, and leave "Add a README" unchecked for now.

2. Prepare your project folder

Put both app files in one folder, along with a README.md describing the project and a requirements.txt listing matplotlib and numpy.

3. Initialize git locally

Open a terminal in that folder and run:

```
git init
git add .
git commit -m "Initial commit: 3D design studio"
```

4. Connect to GitHub and push

Copy the repository URL GitHub shows you after creating it, then run:

```
git branch -M main
git remote add origin https://github.com/your-username/3d-design-studio.git
git push -u origin main
```

5. Verify

Refresh the repository page on GitHub. Your files, including the README, should now be visible.

6. Optional: enable GitHub Pages for the web edition

Since the web edition is a single self-contained HTML file, you can host it for free. In the repository, go to Settings, then Pages, set the source branch to main, and save. Your app will be live at https://your-username.github.io/3d-design-studio/3d-design-studio.html within a few minutes.

Repo checklist

- README.md — what the project is and how to run both editions
- requirements.txt — lists matplotlib and numpy
- LICENSE — MIT is a safe default for personal projects
- .gitignore — excludes __pycache__/, .DS_Store, and editor folders
- 3d-design-studio.html — the web edition
- 3d_design_app.py — the Python edition